

fast_axis_uart.v

AUTHORS

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DATES

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INFORMATION

Brief

Fast UART AXIS core that allows for back to back transmissions.

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fast_axis_uart

```
module fast_axis_uart #(
    parameter
    CLOCK_SPEED
    =
    20000000,
    parameter
    BAUD_RATE
    =
    20000000,
    parameter
    PARITY_TYPE
    =
    0,
    parameter
```

```

STOP_BITS
=
1,
parameter
DATA_BITS
=
8,
parameter
RX_BAUD_DELAY
=
0,
parameter
TX_BAUD_DELAY
=
0
) ( input wire aclk, input wire arstn, output wire parity_err, output wire t

```

AXIS UART, fast simple UART with AXI Streaming interface.

Parameters

CLOCK_SPEED parameter	This is the aclk frequency in Hz
BAUD_RATE parameter	Serial Baud, this can be any value including non-standard.
PARITY_TYPE parameter	Set the parity type, 0 = none, 1 = odd, 2 = even, 3 = mark, 4 = space.
STOP_BITS parameter	Number of stop bits, 1 to crazy non-standard amounts. If you use 0, you will break the protocol.
DATA_BITS parameter	Number of data bits, 1 to 8.
RX_BAUD_DELAY parameter	Delay in rx baud enable. This will delay when we sample a bit (default is midpoint when rx delay is 0).
TX_BAUD_DELAY parameter	Delay in tx baud enable. This will delay the time the bit output starts.

Ports

aclk	Clock for AXIS
arstn	Negative reset for AXIS
parity_err	Indicates error with parity check (active high)
frame_err	Indicates error with frame (active high)
s_axis_tdata	Input data for UART TX.
s_axis_tvalid	When set active high the input data is valid
s_axis_tready	When active high the device is ready for input data.
m_axis_tdata	Output data from UART RX
m_axis_tvalid	When active high the output data is valid
m_axis_tready	When set active high the output device is ready for data.
tx	transmit for UART (output to RX)
rx	receive for UART (input from TX)

INSTANTIATED MODULES

uart_baud_gen_tx

```
mod_clock_ena_gen #(
    CLOCK_SPEED(CLOCK_SPEED),
    DELAY(TX_BAUD_DELAY)
) uart_baud_gen_tx ( .clk(aclk), .rstn(arstn), .start0(1'b1), .clr(s_tx_read
```

Generates TX BAUD rate for UART modules using modulo divide method.

uart_baud_gen_rx

```
mod_clock_ena_gen #(
    CLOCK_SPEED(CLOCK_SPEED),
    DELAY(RX_BAUD_DELAY)
) uart_baud_gen_rx ( .clk(aclk), .rstn(arstn), .start0(1'b0), .clr(rx & r_c
```

Generates RX BAUD rate for UART modules using modulo divide method.

inst_sipo

```
sipo #(
    BUS_WIDTH(4)
) inst_sipo ( .clk(aclk), .rstn(arstn), .ena(uart_ena_rx), .rev(1'b1), .load
```

Captures RX data for uart receive

inst_piso

```
piso #(
    BUS_WIDTH(4),
    DEFAULT_RESET_VAL(1),
    DEFAULT_SHIFT_VAL(1)
) inst_piso ( .clk(aclk), .rstn(arstn), .ena(uart_ena_tx), .rev(1'b1), .load
```

Generates TX data for uart transmit